

Review Form – ICBBE 2026

Paper ID: HD0080

Paper Title: Physiologically Guided R-Centered Deep Learning for ECG Wave Delineation Across Heterogeneous Databases

● Evaluation Summary

Evaluation Criteria	Score (1 = Low / Poor, 5 = High / Excellent)
1. Relevance to Conference Themes	4
2. Clarity and Organization of the Paper	3
3. Scientific Merit / Originality	3
4. Technical Quality	3
5. Figures and Tables	3
6. Language and Style	3
7. References and Background	3

● Overall Recommendation:

- ☐ Accept
☐ Accept with Minor Revision
☒ Accept with Major Revision
☐ Reject

● Comments and Suggestions for Improvement:

This paper addresses an important ECG-analysis problem, and the attempt to evaluate generalization on a second database is a meaningful strength. The current version, however, does not provide enough methodological detail to reproduce the work, and the experimental comparisons do not yet isolate the contribution of the proposed physiological guidance. In addition, the claimed delineation improvement is assessed mainly with sample-level classification scores rather than boundary-localization metrics. These are substantial concerns, but they can potentially be addressed using the existing datasets through a transparent, leakage-safe reanalysis. I therefore recommend acceptance with major revision, conditional on the authors completing the analyses described below.

Major concerns

1. The current methodology is insufficient for reproducibility. The author(s) shall report the exact QTDB and LUDB records and leads used, sampling-rate conversion, annotation-to-mask rules, normalization, window length and overlap, handling of record boundaries, class frequencies, and the number of records and windows in every partition. Figure 1 indicates a 512-sample input, but the choice is not defined in the text. Specify all CNN and BiLSTM layers, kernel sizes,

hidden dimensions, dropout, optimizer, learning rate, batch size, stopping criterion, focal-loss parameters, random seeds, and model-selection procedure. The R detector, R audit, POST320 operation, normalized temporal channel, and R-aware decoder also require algorithmic definitions or pseudocode rather than only conceptual diagrams.

2. The author(s) shall clarify the splits and demonstrate that leakage was prevented. All partitions should be created by patient or record before windowing, with every beat, lead, and overlapping window from one record kept in a single partition. The 10% LUDB adaptation subset should likewise be selected at record level before any tuning. State whether LUDB labels or test performance influenced hyperparameter choices. As currently written, R3 appears to be evaluated on all LUDB records while R6 is evaluated on a different remaining subset, so the two reported scores are not directly comparable. Evaluate all relevant models on the same untouched test records and report results across repeated seeds or record-level splits with confidence intervals.
3. If possible, the author(s) shall conduct a matched ablation that isolates each proposed component. Section 3.3 selects R3 as the reference baseline, but R4 applies its hard decoder to R1. R5 then changes both the R-centered representation and post-R context, while R6 adds a temporal channel together with supervised LUDB adaptation. These comparisons cannot identify which component produced the change. Use one fixed baseline, split, and training procedure, then add the R audit, POST320 context, temporal channel, and target adaptation one at a time. The R5 improvement over R3 is only 0.0030 Macro F1, so repeated-run variability and statistical uncertainty are essential before claiming an improvement.
4. The manuscript can be improved with addition of quantitative boundary-level evaluation. Sample-wise Macro F1 is useful, but it is insufficient for a delineation study because long segment overlap can conceal clinically relevant onset and offset errors. Report wave-detection sensitivity and positive predictive value, together with onset and offset error bias and standard deviation in milliseconds for P, QRS, and T waves under a stated matching tolerance. Per-class precision, recall and F1, event-level missed and false waves, and performance by record or pathology should also be supplied. The single examples in Figures 3 and 4 are illustrative but cannot substitute for a complete boundary analysis.
5. The author(s) may strengthen the external-validation protocol. Explain how differences in sampling frequency, lead structure, annotation practice, and morphology between QTDB and LUDB were handled. State the lead-selection rule and whether multiple leads from the same record are treated as correlated observations. Because the framework depends on deterministic R peaks, evaluate the R detector separately on both databases and quantify how missed or extra R peaks affect downstream delineation. A fair comparison with established delineators should use the same records, leads, preprocessing, matching rules, and metrics. Without such a table, the description of the method as competitive is premature.
6. The current experiments show that one rigid decoder harms one baseline, that a later configuration approximately restores performance, and that supervised target-domain adaptation improves performance under a different evaluation protocol. They do not yet establish that progressively refined physiological guidance improves cross-dataset generalization. Rewrite the abstract, discussion, and conclusion after the matched ablation and fixed-test-set analyses are complete. Sharing code, trained weights, record lists, and split files would materially improve the credibility and usefulness of the work.

Additional comments

- Use one database name consistently (LUDB rather than alternating with 'LU Database') and state whether POST320 means 320 samples or 320 milliseconds at the specified sampling rate.
- Clarify all class colors and segment labels in Figures 3 and 4, and explain how the displayed beats were selected.